

# ROHAN LAL DAS

## Electrical Engineer | Machine Learning & IoT

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### Summary

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Electrical Engineering graduate with comprehensive expertise in smart industrial automation, embedded systems, and deep learning optimization. Experienced in architecting end-to-end cloud-distributed AI pipelines, programming concurrent microcontroller setups, and managing closed-loop feedback systems. Highly skilled at engineering resource-optimized solutions that bridge physical power hardware with high-accuracy predictive intelligence.

### Skills

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**Electrical Engineering & Automation:** Closed-Loop PI Control, H-Bridge Circuit Design, MATLAB & Simulink Modeling, Proteus Simulation, Hardware Interfacing

**Artificial Intelligence & Vision:** Computer Vision, YOLOv4, SqueezeNet, Neural Collaborative Filtering (NCF), CNN Classification, Transformer (BERT)

**Embedded Systems & IoT:** ESP32 Core Programming, Arduino, Firebase Realtime Database, Wireless Sensor Networks (WebSockets/REST)

**Full-Stack & Cloud Development:** Python, FastAPI, Next.js, React, Tailwind CSS, Uvicorn Concurrent Architecture, MongoDB Atlas NoSQL

**Programming & Development Tools:** Python, MATLAB, Google Colab, and Git Version Control

### Experience

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#### AI Data Quality Analyst Intern

Jan 2026 – Apr 2026

*Rooman Technologies Pvt. Ltd. (On-site)*

*Tools Used: FastAPI, Next.js, MongoDB Atlas, Vercel, Render*

- Engineered SenticPulse AI, a cloud-distributed multimodal market intelligence pipeline utilizing asynchronous coroutines to yield sub-200ms processing latencies under live concurrent loads.
- Executed an optimization pivot from heavy Transformer architectures (BERT) to an edge-stabilized VADER engine, cutting active RAM footprint by 98% (under 10MB) to fully circumvent R14 cloud server memory crashes.
- Fused a Neural Collaborative Filtering (NCF) MLP module to map consumer interaction pathways (Hit Ratio HR@10 of 0.84, MAE < 0.2) with a CNN 20-category image classifier ( $\eta = 91.4\%$ ) to systematically bypass cold-start recommendation anomalies.
- Deployed an end-to-end allocated topology managing over 610,000 product JSON trees via embedded document schemas in MongoDB Atlas with  $O(\log N)$  aggregation queries, secured by a dual-tier Role-Based Access Control (RBAC) layer.

#### Machine Learning Intern

May 2025 – Jul 2025

*National Institute of Technology (NIT), Rourkela, Odisha (On-site)*

*Tools Used: MATLAB, Simulink, Google Colab*

- Architected an end-to-end computerized insulator monitoring pipeline in MATLAB utilizing single-stage YOLOv4 (with a CSPDarknet-53 backbone and PANet neck) to localize outdoor transmission line components, securing 78.4% mAP.
- Integrated a lightweight SqueezeNet classification module employing optimized Fire Modules to categorize detected regions into condition classes (normal, broken sheds/cracks, flashover), achieving an overall classification accuracy of 82.6%.
- Wrapped the trained vision neural weights into custom MATLAB System Objects for live execution, simulating drone-camera deployments over high-voltage networks within a structured Simulink workflow.

#### Student Trainee

May 2024 – Jun 2024

*SAIL, Rourkela Steel Plant (RSP), Odisha (Hybrid)*

*Tools Used: ZOOM for meeting*

- Analyzed large-scale heavy industrial power distribution network architectures and electrical safety equipment setups across multiple production stages of a silicon steel manufacturing facility.
- Collaborated with cross-functional plant engineering teams to evaluate quality assurance loops and material processing workflows, mastering production line mechanics and real-time operations under high power demands.

### Projects

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#### Smart Hybrid AC–DC Power Supply With Bidirectional Control

– *ESP32, Firebase, PI Loop, Netlify*

- Assembled a fully integrated hardware prototype featuring an AC-DC rectifier ( $\pm 5\%$  regulation threshold), a 220V AC RMS H-bridge inverter topology, and a non-isolated bidirectional DC-DC buck/boost converter stage.

- Programmed a centralized ESP32 control core utilizing closed-loop Proportional-Integral (PI) feedback controls to dictate smooth power flow switching between Grid-to-Vehicle (G2V) charging and Vehicle-to-Grid (V2G) injection modes.
- Wired end-to-end IoT tracking protocols over WebSockets to sync live parameter matrices (voltage, current spikes, and battery SoC updates) onto a cloud-synchronized web/mobile dashboard with under 0.2s command latency.

### ML-Based Solar Panel Dust Detecting & Cleaning System

– *MobileNetV3, ESP32, Firebase, L298N*

- Built an automated solar panel cleaner using a 12V Li-ion battery, L298N motor driver, and a 3D-printed tracking rail for mechanical wiper sweeps.
- Compared a Custom CNN (77% accuracy) and a YOLO-style CNN (82% accuracy) against an improved MobileNetV3 model to find the best deployment option.
- Applied Transfer Learning on MobileNetV3 to detect surface dust with 99% accuracy and a 0.99 F1-score, eliminating accidental cleaning triggers.
- Programmed an ESP32 microcontroller to start mechanical washing cycles and run post-sweep cleanliness checks once dust levels cross a 40% threshold.
- Linked the system to Firebase to stream real-time voltage data (230V), PWM motor signals (234), and live efficiency logs onto a custom web dashboard.

### Automatic Car Parking System

– *Arduino, C++, Servo Architecture*

- Systemized real-time parking slot monitoring using Arduino, IR sensors, servo motors, and display.
- Guided vehicle positioning for smart parking in malls, offices, and garages.

### Awards & Certifications

- **1st place in Interdepartmental Project Exhibition – 2025:** Recognized for team collaboration and problem-solving in developing a Smart Hybrid AC-DC Power Supply system.
- **All India Naukri Campus Aptitude Test (AINCAT 2026):** Achieved a national rank of 15k, placing in the top 15% out of over 100,000+ appeared engineering candidates across India in an assessment evaluating advanced analytical and technical problem-solving skills.
- **BPUT Hackathon 2025 (Project Finalist):** Selected to present a high-fidelity hardware-software engineering concept; recognized for top-tier architectural design, algorithmic creativity, and deep problem-solving metrics among competitive multi-disciplinary engineering panels.
- **MATLAB Certification (2024 & 25):** Completed fundamentals of MATLAB library and editor through NIELIT.

### Education

#### Bachelor of Technology in Electrical Engineering

Nov 2022 – Jun 2026

*Biju Patnaik University of Technology, Rourkela, Odisha*

*CGPA: 8.49 / 10*

#### Higher Secondary Science

Apr 2020 – Jul 2022

*Ispat Vidya Mandir Higher Secondary School, Sector 19, Rourkela, Odisha*

*Percentage – 76%*